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Predicting respiratory deterioration in ICU patients

Machine Learning and Artificial Intelligence for Early Prediction of Respiratory Deterioration in Mechanically Ventilated ICU Patients: Methods, Performance, and Challenges

Quick Reference

Key Findings Table

Aspect	Summary	Representative Metrics/Findings	Citations
Predictive Methods	Ensemble (GentleBoost, CatBoost), neural networks (MLP, ANN, CNN), gradient boosting (XGBoost, LightGBM), random forest, SVM, time-series models (LSTM, SREM), hybrid/dynamic models	AUROC 0.79–0.94 (best models); accuracy up to 98% in specific tasks	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Feature Selection	LASSO, Boruta, SHAP, MRMR, recursive elimination	Improves interpretability and accuracy; enables parsimonious, robust models	17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
Time-Series/Dynamic Modeling	LSTM, SREM, graph convolutional networks, dynamic clustering	Captures evolving physiology, improves early prediction	7 14 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51
Multimodal Data Integration	Combines vital signs, labs, imaging, biosignals, genomics	AUROC often >0.80; enables personalized risk stratification	4 16 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73
Model Performance	High accuracy, AUROC 0.79–0.94, sensitivity/specificity >0.80 in best models	External validation: AUC drop ~0.037; calibration often good but sometimes suboptimal	1 2 4 6 7 11 12 14 29 30 33 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

Aspect	Summary	Representative Metrics/Findings	Citations
Limitations	Retrospective/single-center studies, limited external validation, workflow integration, interpretability, bias, ethical concerns	High risk of bias, lack of code/data sharing, limited real-world impact studies	4 14 33 42 95 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120

Direct Answer

Machine learning and artificial intelligence can predict early respiratory deterioration in mechanically ventilated ICU patients by leveraging comprehensive data including vital signs (e.g., respiratory rate, heart rate, oxygen saturation), laboratory results, arterial blood gas parameters, and ventilator-derived time-series data. Various methods have been used including ensemble methods (such as GentleBoost-Ensemble and CatBoost), neural networks (e.g., multilayer perceptron, convolutional neural networks), gradient boosting machines (XGBoost, LightGBM), random forest models, and hybrid dynamic models that incorporate time-series analyses. These models have achieved high predictive performances with AUROCs typically ranging from 0.79 to 0.94, and in some cases, even higher accuracy in specific applications like predicting extubation success or ICU mortality. However, limitations remain, such as predominantly retrospective study designs, limited external and prospective validation, integration challenges into clinical workflows, issues with model interpretability and explainability, and ethical and bias concerns that need to be addressed before widespread clinical adoption. 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 30 33 42 74 75 76 77 78 79 80 81 82 101 102 103 104

Study Scope

- **Time Period:** 2015–2024 (majority of studies published 2018–2024)
- **Disciplines:** Critical care medicine, biomedical informatics, machine learning, clinical epidemiology
- **Methods:** Systematic reviews, meta-analyses, retrospective and prospective cohort studies, multicenter trials, algorithmic benchmarking, external validation studies

Assumptions & Limitations

- Most evidence is derived from retrospective, single-center studies with limited external validation.
- Data and code availability is low, impeding reproducibility.
- High risk of bias and confounding, especially due to lack of sociodemographic data and inconsistent calibration.
- Clinical impact and workflow integration are rarely assessed in real-world settings.
- Ethical, fairness, and explainability concerns remain under-addressed.

Suggested Further Research

- Prospective, multicenter clinical trials with standardized protocols and reporting.
- Integration of multimodal data (biosignals, imaging, genomics) for personalized prediction.
- Development and adoption of explainable AI frameworks tailored to ICU workflows.
- Systematic assessment of clinical impact, workflow integration, and user trust.
- Addressing bias, fairness, and ethical considerations through inclusive data practices and transparent algorithms.

1. Introduction

Background and Clinical Relevance

Early respiratory deterioration in mechanically ventilated ICU patients is a critical event associated with increased morbidity, mortality, and resource utilization. Timely identification of patients at risk enables proactive interventions, potentially improving outcomes and reducing complications such as ventilator-associated events, prolonged mechanical ventilation, and ICU-acquired weakness. Traditional risk stratification tools (e.g., APACHE II, SOFA) offer limited predictive accuracy and are often static, failing to capture the dynamic and heterogeneous nature of critical illness [1] [101].

The emergence of machine learning (ML) and artificial intelligence (AI) in critical care has enabled the development of predictive models that leverage high-dimensional, multimodal data—including vital signs, laboratory results, ventilator waveforms, and imaging—to provide individualized, real-time risk assessments. These models promise to augment clinical decision-making, support early intervention, and facilitate personalized management of mechanically ventilated patients [1] [101].

Scope and Structure of the Report

This report synthesizes current evidence on ML and AI methods for early prediction of respiratory deterioration in mechanically ventilated ICU patients, focusing on:

- Predictive methods and algorithms
- Model performance and accuracy
- Limitations and challenges
- Research gaps and future directions

The report bridges theoretical foundations with practical insights, emphasizing methodological rigor, clinical applicability, and translational challenges [1] [98].

2. Theoretical Frameworks

Overview of Machine Learning and AI Models

A wide array of ML and AI algorithms have been applied to predict early respiratory deterioration and related outcomes in ICU patients:

- **Ensemble Methods:** GentleBoost-Ensemble, CatBoost, XGBoost, LightGBM—offer high accuracy and robustness, often outperforming traditional scoring systems [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16].
- **Neural Networks:** Multilayer perceptron (MLP), artificial neural networks (ANN), convolutional neural networks (CNN)—capture complex nonlinear relationships and are effective for high-dimensional and time-series data [4] [12] [13] [14].
- **Random Forest and SVM:** Provide strong baseline performance, especially with structured clinical data [7] [11].
- **Hybrid/Dynamic Models:** Two-stage models, autoencoder-based clustering, and dynamic phenotyping approaches support ventilator weaning and personalized management [9].
- **Time-Series Models:** LSTM, SREM, graph convolutional networks—capture temporal trends and evolving patient physiology, improving early detection [33] [41] [44] [46].

These models are typically trained on combinations of vital signs (respiratory rate, heart rate, SpO2), laboratory results (ABG, inflammatory markers), ventilator parameters, and sometimes imaging or biosignal data [1] [2] [3] [4] [5] [6] [7] [8] [9] [12] [13] [14] [15] [16].

Feature Selection and Interpretability

Feature selection is critical for both model performance and clinical interpretability:

- **LASSO Regression:** Selects a parsimonious set of clinically relevant variables, improving accuracy and interpretability [17] [18] [19] [25] [27] [28] [29] [31].
- **Boruta Algorithm:** Identifies all relevant features, often used in combination with LASSO for robust selection [22] [24] [30] [32].
- **SHAP (SHapley Additive exPlanations):** Provides post-hoc interpretability, quantifying feature importance and explaining individual predictions, which enhances clinician trust [21] [26] [27] [28] [29] [30] [31].

These methods enable the development of models that are not only accurate but also transparent and clinically actionable.

Time-Series and Dynamic Modeling Approaches

Dynamic and time-series modeling is increasingly recognized as essential for early prediction:

- **LSTM and SREM:** Capture temporal dependencies and evolving risk, outperforming static models [33] [41] [46] [47] [50].

- **Graph Convolutional Networks:** Model complex relationships among time-varying variables, improving prediction of interventions like mechanical ventilation [44].
- **Dynamic Clustering and Phenotyping:** Identify patient subgroups with distinct risk trajectories, supporting personalized management [9].

These approaches facilitate real-time monitoring and proactive clinical decision-making.

Multimodal Data Integration

Integrating diverse data types enhances prediction:

- **Vital Signs + Labs + Imaging + Biosignals + Genomics:** Multimodal models consistently outperform unimodal approaches, achieving AUROCs >0.80 and enabling individualized risk stratification [4] [16] [52] [53] [54] [55] [56] [57] [58] [59] [60] [61] [62] [63] [64] [65] [66] [67] [68] [69] [70] [71] [72] [73].
- **Personalized Prediction:** Multi-omics and radiomics integration enables identification of novel biomarkers and phenotypes, supporting precision medicine [54] [60] [64] [67].

Synthesis: Theoretical advances in ML/AI, feature selection, dynamic modeling, and multimodal integration underpin the development of high-performing, interpretable, and clinically relevant prediction models for respiratory deterioration in ICU patients.

3. Methods & Data Transparency

Data Sources and Study Designs

- **Retrospective Cohorts:** Most studies use retrospective EHR data from single or multiple ICUs, with variable sample sizes and data quality [101] [102].
- **Prospective/Multicenter Trials:** Limited but increasing; essential for external validation and real-world assessment [77] [121] [122] [123].
- **Data Types:** Structured (vital signs, labs, ventilator settings), unstructured (clinical notes), imaging, biosignals, and omics data [4] [16] [52] [53] [66].

Model Development and Validation

- **Internal Validation:** Cross-validation, bootstrapping, and split-sample approaches are common.
- **External Validation:** Performed in a minority of studies; reveals modest performance drop (AUC reduction ~0.037) [87].
- **Calibration and Decision-Curve Analysis:** Increasingly reported, supporting clinical utility assessment [6] [30] [124] [125] [126] [127] [128] [129].

Transparency and Reproducibility

- **Data/Code Availability:** Severely limited; hinders reproducibility and independent validation [101].
- **Reporting Standards:** Adherence to TRIPOD, PROBAST, SALIENT, and CHEERS-AI is low but improving [130] [131] [132] [133].

Synthesis: Methodological rigor, transparent reporting, and external validation are essential for reliable, generalizable AI models in ICU respiratory care.

4. Critical Analysis of Findings

Model Accuracy and Performance

Reported Performance Metrics

- **Best Models:** AUROC 0.79–0.94; accuracy up to 98% in specific tasks (e.g., prolonged ventilation, extubation success) [1] [2] [6] [7] [11] [12] [14] [30] [33] [74] [75] [76] [77] [78] [79] [80] [81] [82].
- **Sensitivity/Specificity:** Often >0.80 in high-performing models; some models achieve >0.90 [1] [7] [11] [134] [135] [136].
- **Calibration:** Generally good, but some models require local recalibration for new populations [6] [30] [124] [125] [126] [127] [128] [129].

Comparative Performance Across Subgroups

- **ARDS, COVID-19, Trauma, Pediatric:** AI models outperform traditional scores in most subgroups; best performance in moderate/severe ARDS and specific trauma populations [72] [75] [102] [106] [134] [135] [136] [137] [138] [139] [140] [141] [142] [143] [144] [145] [146].
- **Key Predictors:** Heart rate, respiratory rate, SpO2, age, comorbidities, ABG parameters, and ventilator settings [147] [148] [149] [150] [151] [152] [153] [154] [155].

External Validation and Generalizability

- **Performance Drop:** AUC reduction ~0.037 on external datasets; highlights overfitting and need for multicenter validation [87].
- **Generalizability:** Improved by multicenter data pooling and parsimonious models; still limited by data heterogeneity and lack of prospective trials [29] [83] [84] [86] [90] [93].

Calibration and Clinical Utility

- **Decision-Curve Analysis:** Supports net clinical benefit of AI models over treat-all or treat-none strategies [6] [30] [124] [125] [126] [127] [128] [129].
- **Clinical Impact:** Rarely assessed in real-world settings; evidence of improved workflow and patient outcomes is limited [102] [103].

Synthesis: AI models for respiratory deterioration prediction demonstrate high accuracy and clinical promise, but generalizability, calibration, and real-world impact require further validation.

5. Real-world Implications

Clinical Integration and Workflow

- **Barriers:** Limited prospective validation, workflow disruption, alert fatigue, lack of clinician trust, and model opacity [101] [102] [103] [156] [157] [158] [159] [160] [161] [162].
- **Explainability:** SHAP, LIME, and inherently interpretable models (glass-box) are being developed to enhance transparency and clinician trust [27] [100] [163] [164].
- **User-Centered Design:** Layered explanation interfaces tailored to clinician roles (nurses vs. physicians) improve usability and trust [163] [165] [166] [167].

Ethical, Bias, and Fairness Considerations

- **Bias Sources:** Data quality, missing sociodemographic variables, algorithmic bias, confounding by indication [115] [116] [117] [118] [120].
- **Fairness:** Fragmented metrics, lack of subgroup analysis, and insufficient clinical validation challenge equity [114] [116] [117].
- **Ethical Frameworks:** Emphasize autonomy, privacy, accountability, and equity; require robust governance and continuous monitoring [156] [168] [169] [170] [171] [172] [173].

Implementation Challenges and Solutions

- **Technical:** Data quality, real-time integration, EHR interoperability, performance drift [102] [130] [157] [158].
- **Organizational:** Multidisciplinary collaboration, clinician training, regulatory adaptation [156] [174] [175] [176].
- **Lifecycle-Oriented Approaches:** Local readiness assessment, silent-mode calibration, workflow-integrated deployment, continuous validation, and algorithmic stewardship are recommended [158] [177].

Synthesis: Real-world adoption of AI models for respiratory deterioration prediction hinges on explainability, fairness, workflow integration, and robust governance.

6. Future Research Directions

Current Research Gaps

- **Prospective Multicenter Trials:** Scarce; most studies are retrospective and single-center, limiting generalizability [98] [101] [102] [103].

- **Clinical Impact Assessment:** Few studies evaluate effects on workflow, patient outcomes, or clinician behavior [102] [103].
- **Data/Code Availability:** Low, impeding reproducibility and independent validation [101] [102].
- **Bias and Fairness:** Underreported; need for inclusive data and fairness-aware algorithms [115] [116] [117] [118] [120].

Multi-Center Prospective Trials

- **Status:** Limited but growing; e.g., MEMORIAL, MOTIVATE-ICU, PRoVENT-iMIC studies [121] [122] [123].
- **Needs:** Standardized protocols, diverse populations, real-time implementation, and impact assessment [72] [77] [105] [108] [121] [122] [123] [178] [179] [180] [181] [182] [183].

Advances in Biomarkers and Multimodal Data Integration

- **Multi-omics, Radiomics, Biosignals:** Enhance personalized prediction and risk stratification [53] [54] [55] [56] [57] [58] [59] [60] [61] [62] [63] [64] [65] [66] [67] [68] [69] [70] [71] [72] [73].
- **Personalized Management:** Integration of clinical, organizational, and demographic factors is needed for precision medicine [14] [89] [104].

Explainable AI Frameworks and Reporting Standards

- **Frameworks:** SALIENT, TRIPOD, PROBAST, CHEERS-AI, GRAITE-USRA—support transparent, reproducible, and ethical AI research [16] [130] [131] [132] [133] [184] [185] [186] [187] [188] [189].
- **Recommendations:** Adherence to reporting standards, prospective validation, and multidisciplinary collaboration are essential [132] [186] [189].

Implementation Challenges and Solutions

- **Lifecycle-Oriented Implementation:** Local readiness, silent-mode calibration, workflow integration, continuous validation, and algorithmic stewardship [158] [177].
- **User Involvement:** Active engagement of clinicians, nurses, and other stakeholders in design and deployment [130] [158] [175].
- **Technological Solutions:** AIoT-enabled data fusion, real-time predictive platforms, and EHR integration [40] [190].

Synthesis: Addressing research gaps through prospective trials, multimodal integration, explainable AI, and robust implementation strategies is critical for translating AI prediction models into effective, trustworthy clinical tools in ICU respiratory care.

Conclusion

Machine learning and AI have demonstrated substantial promise in predicting early respiratory deterioration in mechanically ventilated ICU patients, leveraging multimodal data and advanced modeling techniques to achieve high predictive accuracy. However, significant challenges remain in terms of generalizability, external validation, workflow integration, interpretability, and ethical considerations. Future research must prioritize prospective multicenter trials, explainable AI frameworks, and real-world impact assessment to ensure safe, equitable, and effective clinical adoption. Multidisciplinary collaboration and adherence to rigorous methodological standards will be essential to realize the full potential of AI-driven prediction in critical care.

References

1. ML-Driven Prediction of Invasive Mechanical Ventilation in Sepsis: Insights from a tertiary care hospital in India Mendiratta, S., Mukkelli, V.G., Kayal, E.B., (...), Mehndiratta, A. Proceedings of the 2025 International Conference on Artificial Intelligence and Emerging Technology, Global AI Summit 2025, 2025 https://www.scopus.com/pages/publications/105035301795?origin=scopusAI

2. Machine learning models to predict 30-day mortality in mechanically ventilated patients Kim, J.H., Kwon, Y.S., Baek, M.S. Journal of Clinical Medicine, 2021 https://www.scopus.com/pages/publications/85111938987?origin=scopusAI

3. Efficient Machine Learning Model Deployment in Clinical Decision Support Systems Alvarez, R., Chen, J., Nguyen, Q.T. Canadian Conference on Electrical and Computer Engineering, 2024 https://www.scopus.com/pages/publications/85204950069?origin=scopusAI

4. Machine learning-based prediction of intensive care unit admission in COVID-19 patients presenting with mild respiratory failure Ceylan, B., Ceylan, Ş., Olmuşçelik, O., (...), Mert, A. Frontiers in Medicine, 2026 https://www.scopus.com/pages/publications/105031638416?origin=scopusAI

5. Dynamic prediction of life-threatening events for patients in intensive care unit Hu, J., Kang, X.-H., Xu, F.-F., (...), Weng, L. BMC Medical Informatics and Decision Making, 2022 https://www.scopus.com/pages/publications/85140294034?origin=scopusAI

6. Machine Learning-Based Model for Predicting Prolonged Mechanical Ventilation in Patients with Congestive Heart Failure Li, L., Tu, B., Xiong, Y., (...), Yao, Y. Cardiovascular Drugs and Therapy, 2024 https://www.scopus.com/pages/publications/85141962826?origin=scopusAI

7. Developing a machine-learning model for real-time prediction of successful extubation in mechanically ventilated patients using time-series ventilator-derived parameters Huang, K.-Y., Hsu, Y.-L., Chen, H.-C., (...), Hou, M.-H. Frontiers in Medicine, 2023 https://www.scopus.com/pages/publications/85159952058?origin=scopusAI

8. Prediction of hospital mortality in mechanically ventilated patients with congestive heart failure using machine learning approaches Li, L., Zhang, Z., Xiong, Y., (...), Yao, Y. International Journal of Cardiology, 2022 https://www.scopus.com/pages/publications/85129237956?origin=scopusAI

9. Dynamic Clustering-Based Clinical Decision Support Approach for Patient Phenotyping in Mechanical Ventilation Tasci, S.K., Ozturk, M. SIU 2026 - 34th Signal Processing and Communications Applications Conference, Conference Proceedings, 2026 https://www.scopus.com/pages/publications/105048611142?origin=scopusAI

10. Performance of Machine Learning Algorithms in Predicting Prolonged Mechanical Ventilation in Patients with Blunt Chest Trauma Chen, Y., Lu, X., Zhang, Y., (...), Zhang, B. Therapeutics and Clinical Risk Management, 2024 https://www.scopus.com/pages/publications/85205250661?origin=scopusAI

11. A Machine Learning decision-making tool for extubation in Intensive Care Unit patients Fabregat, A., Magret, M., Ferré, J.A., (...), Bodí, M. Computer Methods and Programs in Biomedicine, 2021 https://www.scopus.com/pages/publications/85096840678?origin=scopusAI

12. Prediction prolonged mechanical ventilation in trauma patients of the intensive care unit according to initial medical factors: a machine learning approach Vali, M., Paydar, S., Seif, M., (...), Ghaem, H. Scientific Reports, 2023 https://www.scopus.com/pages/publications/85152352122?origin=scopusAI

13. Development of a Machine Learning Model for Predicting Weaning Outcomes Based Solely on Continuous Ventilator Parameters during Spontaneous Breathing Trials Park, J.E., Kim, D.Y., Park, J.W., (...), Chung, W.Y. Bioengineering, 2023 https://www.scopus.com/pages/publications/85175168002?origin=scopusAI

14. Predicting Outcome and Duration of Mechanical Ventilation in Acute Hypoxemic Respiratory Failure: The PREMIER Study Villar, J., González-Martín, J.M., Fernández, C., (...), Szakmany, T. Journal of Clinical Medicine, 2025 https://www.scopus.com/pages/publications/105022907722?origin=scopusAI

15. Individualised treatment effects of enhanced early mobilisation in mechanically ventilated patients: a secondary analysis of the TEAM trial Hodgson, C.L., Spicer, A.B., Broadley, T., (...), Churpek, M.M. Intensive Care Medicine, 2026 https://www.scopus.com/pages/publications/105022843380?origin=scopusAI

16. AI in Critical Care and Mechanical Ventilation Predictive Models: Closed-Loop Systems, ARDS Intelligence, and Multimodal ICU Decision Support Kamdar, A., Deshmukh, V., Chunkhare, M. Opportunities, Risks, and Future Directions of AI in Pulmonary Medicine, 2026 https://www.scopus.com/pages/publications/105044197220?origin=scopusAI

17. LASSO-Based Identification of Risk Factors and Development of a Prediction Model for Sepsis Patients Hong, C., Xiong, Y., Xia, J., (...), Chen, Y. Therapeutics and Clinical Risk Management, 2024 https://www.scopus.com/pages/publications/85185144858?origin=scopusAI

18. Significant clinical factors for mortality prediction in ICU sepsis patients: A machine learning approach Santos, C., Garzón, J.X.R., Pertuz, S.D., Fajardo, C.A. Smart Health, 2025 https://www.scopus.com/pages/publications/105018080971?origin=scopusAI

19. Interpretable machine learning based on the Charlson comorbidity index predicts 28-day mortality in acute hypercapnic respiratory failure Lu, C., Lin, J., Yue, Y., (...), Zhang, G. Scientific Reports, 2026 https://www.scopus.com/pages/publications/105028414979?origin=scopusAI

20. Patient Management Assisted by a Neural Network Reduces Mortality in an Intermediate Care Unit Heili-Frades, S., Minguez, P., Mahillo Fernández, I., (...), Peces-Barba Romero, G. Archivos de Bronconeumologia, 2020 https://www.scopus.com/pages/publications/85078175979?origin=scopusAI

21. Machine learning-based prognostic modeling using clinical data and quantitative radiomic features from chest CT images in COVID-19 patients Shiri, I., Sorouri, M., Geramifar, P., (...), Zaidi, H. Computers in Biology and Medicine, 2021 https://www.scopus.com/pages/publications/85101960754?origin=scopusAI

22. Development and validation of a prognostic model for acute respiratory distress syndrome in critically Ill patients with intra-abdominal sepsis: a multicenter cohort study Li, W., Li, Y., Wang, Y., (...), Yu, X. Frontiers in Medicine, 2026 https://www.scopus.com/pages/publications/105033872939?origin=scopusAI

23. Proposing an Intelligent Monitoring System for Early Prediction of Need for Intubation among COVID-19 Hospitalized Patients Afrash, M.R., Kazemi-Arpanahi, H., Nopour, R., (...), Shanbehzadeh, M. Journal of Environmental Health and Sustainable Development, 2022 https://www.scopus.com/pages/publications/85141180997?origin=scopusAI

24. Study on a 28-day prognostic model for ICU patients with AB infection based on machine learning Zhang, Y., Yang, K., Huang, J., (...), Xiao, L. Medicine (United States), 2025 https://www.scopus.com/pages/publications/105024983638?origin=scopusAI

25. Comparison of machine learning and nomogram to predict 30-day in-hospital mortality in patients with acute myocardial infarction combined with cardiogenic shock: a retrospective study based on the eICU-CRD and MIMIC-IV databases Shen, C., Wang, S., Huo, R., (...), Yang, S. BMC Cardiovascular Disorders, 2025 https://www.scopus.com/pages/publications/105000396421?origin=scopusAI

26. Multiplex Targeted Proteomic Analysis of Cytokine Ratios for ICU Mortality in Severe COVID-19 Araújo, R., Von Rekowski, C.P., Fonseca, T.A.H., (...), Bento, L. Proteomes, 2025 https://www.scopus.com/pages/publications/105017150242?origin=scopusAI

27. Machine learning-based prediction model for respiratory failure in patients with pneumoconiosis Jiang, Y., Zhang, Q., Wu, D. Chinese Journal of Industrial Hygiene and Occupational Diseases, 2026 https://www.scopus.com/pages/publications/105043502990?origin=scopusAI

28. Dynamic and interpretable deep learning model for predicting respiratory failure following cardiac surgery Xu, M., Liu, H., Dai, A., (...), Si, Y. BMC Anesthesiology, 2025 https://www.scopus.com/pages/publications/105012624827?origin=scopusAI

29. Development of a prediction model for acute respiratory distress syndrome in ICU patients with acute pancreatitis based on machine learning algorithms REN, X., LIU, L., ZHA, J., (...), ZHANG, Y. Chinese Journal of Clinical Research, 2025 https://www.scopus.com/pages/publications/105043092082?origin=scopusAI

30. Construction and validation of a machine learning-based risk prediction model for invasive mechanical ventilation in AECOPD patients complicated with respiratory failure Jiang, X., Li, J., Ju, J., (...), Yang, S. International Journal of Medical Informatics, 2026 https://www.scopus.com/pages/publications/105027386838?origin=scopusAI

31. A prediction model for postoperative respiratory failure in patients with traumatic cervical spinal cord injury using interpretable machine learning Yu, M., Xing, H., Jiang, T., (...), Li, C. European Spine Journal, 2026 https://www.scopus.com/pages/publications/105023898043?origin=scopusAI

32. Interpretable machine learning-based predictive model for fall risk in older adults receiving maintenance hemodialysis Jiang, M., Zhang, M. Frontiers in Medicine, 2026 https://www.scopus.com/pages/publications/105043872259?origin=scopusAI

33. A multicenter machine learning model for predicting ICU mortality in mechanically ventilated patients: development and external validation Zhang, Y., Chen, X., Tian, X., Zhao, H. BMC Medical Informatics and Decision Making, 2026 https://www.scopus.com/pages/publications/105052257070?origin=scopusAI

34. Predicting mechanically ventilated patients future respiratory system elastance – A stochastic modelling approach Ang, C.Y.S., Chiew, Y.S., Wang, X., (...), Chase, J.G. Computers in Biology and Medicine, 2022 https://www.scopus.com/pages/publications/85141474459?origin=scopusAI

35. Development and validation of a mechanical power-oriented prediction model of weaning failure in mechanically ventilated patients: a retrospective cohort study Yan, Y., Luo, J., Wang, Y., (...), Li, X. BMJ Open , 2022 https://www.scopus.com/pages/publications/85144212582?origin=scopusAI

36. Assessing the asynchrony event based on the ventilation mode for mechanically ventilated patients in ICU Sauki, N.S.M., Damanhuri, N.S., Othman, N.A., (...), Nor, M.B.M. Bioengineering, 2021 https://www.scopus.com/pages/publications/85121766726?origin=scopusAI

37. Individualized Mechanical power-based ventilation strategy for acute respiratory failure formalized by finite mixture modeling and dynamic treatment regimen Hong, Y., Chen, L., Pan, Q., (...), Zhang, Z. EClinicalMedicine, 2021 https://www.scopus.com/pages/publications/85106921089?origin=scopusAI

38. Mechanical power normalized to dynamic lung compliance as a predictor of weaning and prognosis in mechanically ventilated patients: a retrospective cohort study Yan, Y., Gu, X., Xu, H., (...), Xie, Y. BMC Anesthesiology, 2025 https://www.scopus.com/pages/publications/105018294928?origin=scopusAI

39. Estimation of Asynchrony Events with Negative Elastance in Spontaneously Breathing Mechanically Ventilated Patients in ICU Sauki, N.S.M., Damanhuri, N.S., Othman, N.A., (...), Ralib, A.M. 2022 8th International Conference on Control, Decision and Information Technologies, CoDIT 2022, 2022 https://www.scopus.com/pages/publications/85134323233?origin=scopusAI

40. Advancing ICU patient care with a Real-Time predictive model for mechanical Power to mitigate VILI Ruiz-Botella, M., Manrique, S., Gomez, J., Bodí, M. International Journal of Medical Informatics, 2024 https://www.scopus.com/pages/publications/85195192080?origin=scopusAI

41. Multitask multilayer-prediction model for predicting mechanical ventilation and the associated mortality rate El-Rashidy, N., Tarek, Z., Elshewey, A.M., Shams, M.Y. Neural Computing and Applications, 2025 https://www.scopus.com/pages/publications/85209682937?origin=scopusAI

42. Use of learning approaches to predict clinical deterioration in patients based on various variables: a review of the literature Al-Shwaheen, T.I., Moghbel, M., Hau, Y.W., Ooi, C.Y. Artificial Intelligence Review, 2022 https://www.scopus.com/pages/publications/85105960834?origin=scopusAI

43. Dynamic prediction of adverse outcomes in discharged COPD patients using patient-reported outcomes Jin, J., He, H., Hu, X., (...), Zhang, Y. BMC Medical Research Methodology, 2026 https://www.scopus.com/pages/publications/105047310650?origin=scopusAI

44. Predicting ICU Interventions: A Transparent Decision Support Model Based on Multivariate Time Series Graph Convolutional Neural Network Xu, Z., Guo, J., Qin, L., (...), Li, X. IEEE Journal of Biomedical and Health Informatics, 2024 https://www.scopus.com/pages/publications/85188889492?origin=scopusAI

45. Dynamic and explainable machine learning prediction of mortality in patients in the intensive care unit: a retrospective study of high-frequency data in electronic patient records Thorsen-Meyer, H.-C., Nielsen, A.B., Nielsen, A.P., (...), Perner, A. The Lancet Digital Health, 2020 https://www.scopus.com/pages/publications/85081961002?origin=scopusAI

46. Mortality Prediction in ARDS Patients Using Long Short-Term Memory Networks Anand, S., Chandramohan, K., Hemavathi, S., (...), Swetha, G. 2nd International Conference on Artificial Intelligence for Innovations in Healthcare Industries, ICAIIHI 2025, 2025 https://www.scopus.com/pages/publications/105041766254?origin=scopusAI

47. Dynamic prediction of mortality in COVID-19 patients in the intensive care unit: A retrospective multi-center cohort study Smit, J.M., Krijthe, J.H., Endeman, H., (...), Reinders, M.J.T. Intelligence-Based Medicine, 2022 https://www.scopus.com/pages/publications/85136523482?origin=scopusAI

48. Predicting and Interpreting Clinical Deterioration in the Intensive Care Unit Using Machine Learning and Explainable AI Carvalho Leite Junior, V., Pires, M.G., Bertoni, F.C. Lecture Notes in Computer Science, 2026 https://www.scopus.com/pages/publications/105029562162?origin=scopusAI

49. A pretrain-finetune approach for improving model generalizability in outcome prediction of acute respiratory distress syndrome patients Lin, S., Yang, M., Liu, C., (...), Long, X. International Journal of Medical Informatics, 2024 https://www.scopus.com/pages/publications/85188447854?origin=scopusAI

50. Daily prediction of next-day disease severity in critically ill patients: A prospective cohort study Zhang, W., Cox, E.G.M., Heijkoop, È.R.H., (...), Keus, F. Australian Critical Care, 2025 https://www.scopus.com/pages/publications/105003491606?origin=scopusAI

51. Predicting hospital mortality for intensive care unit patients: Time-series analysis Awad, A., Bader-El-Den, M., McNicholas, J., (...), El-Sonbaty, Y. Health Informatics Journal, 2020 https://www.scopus.com/pages/publications/85070323491?origin=scopusAI

52. Using Artificial Intelligence to Predict Mechanical Ventilation Weaning Success in Patients with Respiratory Failure, Including Those with Acute Respiratory Distress Syndrome Stivi, T., Padawer, D., Dirini, N., (...), Ledot, S. Journal of Clinical Medicine, 2024 https://www.scopus.com/pages/publications/85187897653?origin=scopusAI

53. Enhancing clinical decision support with physiological waveforms — A multimodal benchmark in emergency care Alcaraz, J.M.L., Bouma, H., Strodthoff, N. Computers in Biology and Medicine, 2025 https://www.scopus.com/pages/publications/105003782601?origin=scopusAI

54. Identification of early and intermediate biomarkers for ARDS mortality by multi-omic approaches Liao, S.Y., Casanova, N.G., Bime, C., (...), Garcia, J.G.N. Scientific Reports, 2021 https://www.scopus.com/pages/publications/85115418500?origin=scopusAI

55. Subject-Adaptive Sparse Linear Models for Interpretable Personalized Health Prediction from Multimodal Lifelog Data Bu, D., Han, J., Kwon, S., (...), Lee, J.-S. International Conference on ICT Convergence, 2025 https://www.scopus.com/pages/publications/105035085277?origin=scopusAI

56. A Predictive Machine Learning Tool for Asthma Exacerbations: Results from a 12-Week, Open-Label Study Using an Electronic Multi-Dose Dry Powder Inhaler with Integrated Sensors Lugogo, N.L., Depietro, M., Reich, M., (...), Safioti, G. Journal of Asthma and Allergy, 2022 https://www.scopus.com/pages/publications/85141539535?origin=scopusAI

57. Respiratory Biomarkers: An Opportunity for Patient-Centric Microsampling Approach for Treatment Optimization Khan, A., Stair, J.L., Baker, D., (...), Murnane, D. Clinical and Translational Science, 2026 https://www.scopus.com/pages/publications/105040587479?origin=scopusAI

58. An unsupervised learning approach to identify novel signatures of health and disease from multimodal data Shomorony, I., Cirulli, E.T., Huang, L., (...), Shah, N. Genome Medicine, 2020 https://www.scopus.com/pages/publications/85077765370?origin=scopusAI

59. Feature Importance Guided Network With Missing-Value Learning for Pneumonia Pathogen Classification and Biomarker Discovery Xu, D., Wen, R., Wang, Y., (...), Wang, J. IEEE Transactions on Radiation and Plasma Medical Sciences, 2026 https://www.scopus.com/pages/publications/105017135044?origin=scopusAI

60. Construction of Rheumatoid Arthritis-Associated Interstitial Lung Disease diagnostic model and identification of biomarkers based on a multi-omics integration strategy of machine learning Wu, D., Chen, J., Liang, H., (...), Zou, Q. Clinics, 2026 https://www.scopus.com/pages/publications/105035811064?origin=scopusAI

61. Development and validation of a multimodal risk prediction model for early COPD progression: a retrospective study integrating clinical, functional, and CT radiomic features Zhang, J., Yang, Q., Dai, S., (...), Zou, W. Therapeutic Advances in Respiratory Disease, 2026 https://www.scopus.com/pages/publications/105044736581?origin=scopusAI

62. Predicting mortality with machine learning & biomarkers: A cohort analysis from aerosolized β_2-agonist for treatment of acute lung injury (ALTA) trial Chase, A.M., Almuntashiri, S., Zhang, D., (...), Sikora, A. Physiological Reports, 2026 https://www.scopus.com/pages/publications/105033853809?origin=scopusAI

63. Advancements in biomarkers and machine learning for predicting of bronchopulmonary dysplasia and neonatal respiratory distress syndrome in preterm infants Talebi, H., Dastgheib, S.A., Vafapour, M., (...), Neamatzadeh, H. Frontiers in Pediatrics, 2025 https://www.scopus.com/pages/publications/105004663625?origin=scopusAI

64. Integrative multi-omics analysis unravels the host response landscape and reveals a serum protein panel for early prognosis prediction for ARDS Lin, M., Xu, F., Sun, J., (...), Wang, W. Critical Care, 2024 https://www.scopus.com/pages/publications/85197209153?origin=scopusAI

65. Integrating Genomics and Environmental Data to Predict Climate-Related Respiratory Risk Using Machine Learning Rauf, A.A., Arora, N., Aighobahi, A., (...), Puthiyedth, N. Signals and Communication Technology, 2026 https://www.scopus.com/pages/publications/105049373369?origin=scopusAI

66. Artificial Intelligence in Respiratory Medicine: Applications, Methodological Challenges, and Clinical Translation López-Padilla, D., Lumbreras, S., Agustí, À., (...), Soriano, J.B. Archivos de Bronconeumologia, 2026 https://www.scopus.com/pages/publications/105051175313?origin=scopusAI

67. A Comprehensive Review of Multimodal Large Language Models for Medical Imaging and Omics Data Vavekanand, R. Archives of Computational Methods in Engineering, 2026 https://www.scopus.com/pages/publications/105028891920?origin=scopusAI

68. Integration of Multimodal Large Language Models in Medical Imaging and Omics Data: A Comprehensive Review Vavekanand, R. Frontiers in Biomedical Technologies, 2026 https://www.scopus.com/pages/publications/105031292649?origin=scopusAI

69. Contextualized biomedical language processing enhances ICU survival prediction Chen, R., Cai, Y., Zhang, S., (...), Zhang, X. iScience, 2026 https://www.scopus.com/pages/publications/105025664620?origin=scopusAI

70. ICU mortality prediction via multimodal learning on MIMIC-IV clinical data Falcone, A., Forestiero, A., Malivindi, R., Settino, M. Bioinformatics Advances, 2026 https://www.scopus.com/pages/publications/105047815622?origin=scopusAI

71. Multimodal Model for Predicting Fetal Acidosis in Delivery Room Choi, B., Park, C.E., Park, J.C., (...), Kim, M. Studies in Health Technology and Informatics, 2024 https://www.scopus.com/pages/publications/85202007964?origin=scopusAI

72. Artificial Intelligence-Based Early Prediction of Acute Respiratory Failure in the Emergency Department Using Biosignal and Clinical Data Han, C., Jung, Y.J., Park, J.E., (...), Yoon, D. Yonsei Medical Journal, 2025 https://www.scopus.com/pages/publications/85216720305?origin=scopusAI

73. AI in the Trauma Bay Khalpey, Z., Sucher, J.F. Hot Topics in Acute Care Surgery and Trauma, 2026 https://www.scopus.com/pages/publications/105028533643?origin=scopusAI

74. Machine learning models predicting extubation success in mechanically ventilated patients: a systematic review and meta-analysis Bakos, P., Szabó, B., Laczkó, D., (...), Molnár, Z. Intensive Care Medicine Experimental, 2026 https://www.scopus.com/pages/publications/105044281813?origin=scopusAI

75. Development, deployment, and continuous monitoring of a machine learning model to predict respiratory failure in critically ill patients Lam, J.Y., Lu, X., Shashikumar, S.P., (...), Nemati, S. JAMIA Open, 2024 https://www.scopus.com/pages/publications/85212039163?origin=scopusAI

76. Machine learning for prediction of successful extubation of mechanical ventilated patients in an intensive care unit: A retrospective observational study Otaguro, T., Tanaka, H., Igarashi, Y., (...), Yokota, H. Journal of Nippon Medical School, 2021 https://www.scopus.com/pages/publications/85114100038?origin=scopusAI

77. Derivation and external validation of predictive models for invasive mechanical ventilation in intensive care unit patients with COVID-19 Maia, G., Martins, C.M., Marques, V., (...), Silva, P.L. Annals of Intensive Care, 2024 https://www.scopus.com/pages/publications/85201679027?origin=scopusAI

78. Development and validation of an interpretable machine learning model for predicting reguirement of early invasive mechanical ventilation in patients with acute pancreatitis Wu, X., Sha, T., Hu, H., (...), Chen, Z. Nan Fang

Yi Ke Da Xue Xue Bao / Journal of Southern Medical University, 2026
https://www.scopus.com/pages/publications/105042644966?origin=scopusAI

79. Machine Learning Prediction Models for Mechanically Ventilated Patients: Analyses of the MIMIC-III Database Zhu, Y., Zhang, J., Wang, G., (...), Yu, Q. Frontiers in Medicine, 2021
https://www.scopus.com/pages/publications/85110586059?origin=scopusAI

80. Compact machine learning model for the accurate prediction of first 24-hour survival of mechanically ventilated patients Nguyen, Q.T., Tran, M.P., Prabhakaran, V., (...), Nguyen, G.H. Frontiers in Medicine, 2024
https://www.scopus.com/pages/publications/85197371633?origin=scopusAI

81. Development and Validation of a Deep Learning Classifier Using Chest Radiographs to Predict Extubation Success in Patients Undergoing Invasive Mechanical Ventilation Tandon, P., Nguyen, K.-A.-N., Edalati, M., (...), Kia, A. Bioengineering, 2024 https://www.scopus.com/pages/publications/85197949201?origin=scopusAI

82. Development of an Artificial Intelligence Model to Predict Endotracheal Intubation in Critically Ill Patients in Real Time Moon, D.H., Kim, M., Han, S.-S., (...), Heo, Y. Journal of Clinical Medicine, 2026
https://www.scopus.com/pages/publications/105040240453?origin=scopusAI

83. Predictive model for acute respiratory distress syndrome events in ICU patients in China using machine learning algorithms: A secondary analysis of a cohort study Ding, X.-F., Li, J.-B., Liang, H.-Y., (...), Sun, T.-W. Journal of Translational Medicine, 2019 https://www.scopus.com/pages/publications/85072847670?origin=scopusAI

84. Early Machine Learning-Based Identification of Hospitalized Patients at Low Risk of Respiratory Deterioration or Mortality in Community-Acquired Pneumonia: External Validation of a Multivariable Model Gyimah, C., Pudasaini, P., LeMahieu, A., (...), Odeyemi, Y.E. Infectious Disease Reports, 2026
https://www.scopus.com/pages/publications/105048262879?origin=scopusAI

85. Less is more: Detecting clinical deterioration in the hospital with machine learning using only age, heart rate, and respiratory rate Akel, M.A., Carey, K.A., Winslow, C.J., (...), Edelson, D.P. Resuscitation, 2021
https://www.scopus.com/pages/publications/85114923127?origin=scopusAI

86. Early identification of patients admitted to hospital for covid-19 at risk of clinical deterioration: Model development and multisite external validation study Kamran, F., Tang, S., Otles, E., (...), Wiens, J. The BMJ, 2022
https://www.scopus.com/pages/publications/85124779698?origin=scopusAI

87. External validation of AI-based scoring systems in the ICU: a systematic review and meta-analysis Rockenschaub, P., Akay, E.M., Carlisle, B.G., (...), Madai, V.I. BMC Medical Informatics and Decision Making, 2025 https://www.scopus.com/pages/publications/85214251804?origin=scopusAI

88. Implementation of prognostic machine learning algorithms in paediatric chronic respiratory conditions: A scoping review Filipow, N., Main, E., Sebire, N.J., (...), Stanojevic, S. BMJ Open Respiratory Research, 2022
https://www.scopus.com/pages/publications/85126663060?origin=scopusAI

89. Predicting ICU Mortality in Acute Respiratory Distress Syndrome Patients Using Machine Learning: The Predicting Outcome and STratifiCation of severity in ARDS (POSTCARDS) Study Villar, J., González-Martín, J.M., Hernández-González, J., (...), Kacmarek, R.M. Critical Care Medicine, 2023
https://www.scopus.com/pages/publications/85175713911?origin=scopusAI

90. Development and multi-database validation of interpretable machine learning models for predicting In-Hospital mortality in pneumonia patients: A comprehensive analysis across four healthcare systems Chen, J., Hou, D., Song, Y. Respiratory Research, 2025 https://www.scopus.com/pages/publications/105017751018?origin=scopusAI

91. Multicenter Comparison of Machine Learning Methods and Conventional Regression for Predicting Clinical Deterioration on the Wards Churpek, M.M., Yuen, T.C., Winslow, C., (...), Edelson, D.P. Critical Care Medicine, 2016 https://www.scopus.com/pages/publications/84954349720?origin=scopusAI

92. Comparison of Multimodal Deep Learning Approaches for Predicting Clinical Deterioration in Ward Patients: Observational Cohort Study Kotula, C.A., Martin, J., Carey, K.A., (...), Churpek, M.M. Journal of Medical Internet Research, 2025 https://www.scopus.com/pages/publications/105007870210?origin=scopusAI

93. Multi-source data integration through pooling and transfer learning improves generalizability and specialization of deep learning models for ICU mortality and length of stay prediction: a four-database external validation study Allyn, J., Oliver, M., Cerveau, R., (...), Ferdynus, C. Critical Care, 2026 https://www.scopus.com/pages/publications/105044168509?origin=scopusAI

94. External validation of prognostic models for critically ill patients required substantial sample sizes Peek, N., Arts, D.G.T., Bosman, R.J., (...), de Keizer, N.F. Journal of Clinical Epidemiology, 2007 https://www.scopus.com/pages/publications/34047161057?origin=scopusAI

95. Artificial Intelligence Models for Mortality and Outcome Prediction in Intensive Care Unit Sepsis: A Systematic Review Mazza, G., Neri, G., Mastrangelo, H., (...), Garofalo, E. Journal of Personalized Medicine, 2026 https://www.scopus.com/pages/publications/105049237073?origin=scopusAI

96. Development and external validation of an artificial intelligence model for predicting mortality and prolonged ICU stay in postoperative critically ill patients: a retrospective study Park, D.J., Baik, S.M., Hong, K.S., (...), Lee, J.-M. World Journal of Emergency Surgery, 2025 https://www.scopus.com/pages/publications/105018834375?origin=scopusAI

97. Development and external validation of an ICU mortality prediction model using routinely collected clinical data Zhu, J., Xie, Z., Liu, J., Li, C. Medicine (United States), 2026 https://www.scopus.com/pages/publications/105040750984?origin=scopusAI

98. Integration, challenges, and future of artificial intelligence in critical care medicine: comprehensive applications from predictive models to clinical integration Ji, Q., Wu, Y., Yang, M., (...), Hu, W. Frontiers in Medicine, 2026 https://www.scopus.com/pages/publications/105047096438?origin=scopusAI

99. Deep learning for tubes and lines detection in critical illness: Generalizability and comparison with residents Wongveerasin, P., Tongdee, T., Saiviroonporn, P. European Journal of Radiology Open, 2024 https://www.scopus.com/pages/publications/85199863261?origin=scopusAI

100. Translational Potential and Explainability of Artificial Intelligence-Based Clinical Decision Support for Adults in Intensive Care: A Scoping Review Pranata, Y., Mirwanti, R., Bahtiar, Y.A., Purwanti, H. Journal of Multidisciplinary Healthcare, 2026 https://www.scopus.com/pages/publications/105044973731?origin=scopusAI

101. Artificial intelligence for mechanical ventilation: systematic review of design, reporting standards, and bias Gallifant, J., Zhang, J., del Pilar Arias Lopez, M., (...), Formenti, F. British Journal of Anaesthesia, 2022 https://www.scopus.com/pages/publications/85121143332?origin=scopusAI

102. Artificial intelligence in mechanical ventilation: a narrative review of clinical applications and research gaps Amanguli, M., Yaxiaerjiang, M., Guo, Y.-S., (...), Zhong, M. Journal of Thoracic Disease, 2026 https://www.scopus.com/pages/publications/105049215184?origin=scopusAI

103. Artificial intelligence in the management of patient-ventilator asynchronies: A scoping review Muñoz, J., Ruíz-Cacho, R., Fernández-Araujo, N.J., (...), Muñoz-Visedo, J. Heart and Lung, 2025 https://www.scopus.com/pages/publications/105005576873?origin=scopusAI

104. Predicting the Length of Mechanical Ventilation in Acute Respiratory Disease Syndrome Using Machine Learning: The PIONEER Study Villar, J., González-Martín, J.M., Fernández, C., (...), Steyerberg, E.W. Journal of Clinical Medicine, 2024 https://www.scopus.com/pages/publications/85188999827?origin=scopusAI

105. Can we predict the future of respiratory failure prediction? Pearce, A.K., Nemati, S., Goligher, E.C., (...), Rogers, A. Critical Care, 2025 https://www.scopus.com/pages/publications/105008964171?origin=scopusAI

106. Artificial intelligence in predicting anesthetic complications: current techniques, clinical applications, and limitations Mohammadi, A. International Journal of Medical Informatics, 2026 https://www.scopus.com/pages/publications/105042053508?origin=scopusAI

107. Current Perspectives on the Use of Artificial Intelligence in Critical Patient Safety Barea Mendoza, J.A., Valiente Fernandez, M., Pardo Fernandez, A., Gómez Álvarez, J. Medicina Intensiva, 2025 https://www.scopus.com/pages/publications/85190748473?origin=scopusAI

108. Artificial intelligence assisted telemedicine, clinical decision support for anesthesia and critical care in intensive care units: a scoping review Yang, Q., Li, M., Lei, Y. BMC Anesthesiology, 2026 https://www.scopus.com/pages/publications/105045382781?origin=scopusAI

109. Machine learning-based prediction of early invasive mechanical ventilation in ICU patients with pneumonia: Development and external validation Lu, Y., Chen, H., Li, M., (...), Lv, H. International Journal of Medical Informatics, 2026 https://www.scopus.com/pages/publications/105041309729?origin=scopusAI

110. The value of machine and deep learning in management of critically ill patients: An umbrella review Tungushpayev, M., Suleimenova, D., Sarria-Santamerra, A., (...), Viderman, D. International Journal of Medical Informatics, 2025 https://www.scopus.com/pages/publications/105012924970?origin=scopusAI

111. Causal counterfactual simulation for treatment decisions in multimodal lung disease data Zhu, Y., Zhang, L., Sainsbury, C., (...), Ye, X. Computers in Biology and Medicine, 2026 https://www.scopus.com/pages/publications/105041339218?origin=scopusAI

112. Rule-based cohort definitions for acute respiratory failure: Electronic phenotyping algorithm Essay, P., Mosier, J., Subbian, V. JMIR Medical Informatics, 2020 https://www.scopus.com/pages/publications/85089088386?origin=scopusAI

113. Validation of a machine learning model for predicting early deterioration in the emergency department Lee, Y.R., Ruffolo, I., Mashouri, P., (...), Ben-Yakov, M. American Journal of Emergency Medicine, 2026 https://www.scopus.com/pages/publications/105039892047?origin=scopusAI

114. Equitable early prediction of icu length of stay from first-day clinical data Jeyaraj, H.R., Abakasanga, E., Webber, T., Dai, D. Studies in Health Technology and Informatics, 2026 https://www.scopus.com/pages/publications/105044855446?origin=scopusAI

115. Introducing the Team Card: Enhancing governance for medical Artificial Intelligence (AI) systems in the age of complexity Modise, L.M., Avanaki, M.A., Ameen, S., (...), Comeau, D.S. PLOS Digital Health, 2025 https://www.scopus.com/pages/publications/86000324981?origin=scopusAI

116. What's fair is ⋯ fair? Presenting JustEFAB, an ethical framework for operationalizing medical ethics and social justice in the integration of clinical machine learning Mccradden, M., Odusi, O., Joshi, S., (...), Goldenberg, A. ACM International Conference Proceeding Series, 2023 https://www.scopus.com/pages/publications/85163624761?origin=scopusAI

117. Critical appraisal of fairness metrics for artificial intelligence-based clinical prediction models: a scoping review Matos, J., Van Calster, B., Celi, L.A., (...), Collins, G.S. The Lancet Digital Health, 2026 https://www.scopus.com/pages/publications/105040514049?origin=scopusAI

118. Availability of information needed to evaluate algorithmic fairness — A systematic review of publicly accessible critical care databases Fong, N., Langnas, E., Law, T., (...), Pirracchio, R. Anaesthesia Critical Care and Pain Medicine, 2023 https://www.scopus.com/pages/publications/85162188962?origin=scopusAI

119. Natural language processing techniques to detect delirium in hospitalized patients from clinical notes: a systematic review Shankar, R., Kannan, J., Tan, Y.H., Xu, Q. npj Digital Medicine, 2025 https://www.scopus.com/pages/publications/105022600039?origin=scopusAI

120. The Sociodemographic Biases in Machine Learning Algorithms: A Biomedical Informatics Perspective Franklin, G., Stephens, R., Piracha, M., (...), Elkin, P.L. Life, 2024 https://www.scopus.com/pages/publications/85197252425?origin=scopusAI

121. Early Prediction of ICU Mortality in Patients with Acute Hypoxemic Respiratory Failure Using Machine Learning: The MEMORIAL Study Villar, J., González-Martín, J.M., Fernández, C., (...), Szakmany, T. Journal of Clinical Medicine, 2025 https://www.scopus.com/pages/publications/86000645503?origin=scopusAI

122. Potentially MOdifiable factors To ImproVe outcomes of mechanically Ventilated patients in a low-income country Intensive Care Units (MOTIVATE-ICU): rationale and protocol for a registry-embedded prospective observational study Sendagire, C., Pisani, L., Nuwagira, A., (...), Salluh, J.I.F. Critical Care Science, 2025 https://www.scopus.com/pages/publications/105019678646?origin=scopusAI

123. PRactice of VENTilation in Middle-Income Countries (PRoVENT-iMIC): Rationale and protocol for a prospective international multicentre observational study in intensive care units in Asia Pisani, L., Algera, A.G., Serpa Neto, A., (...), Schultz, M.J. BMJ Open, 2018 https://www.scopus.com/pages/publications/85057037600?origin=scopusAI

124. Explainable SHAP-XGBoost models for pressure injuries among patients requiring with mechanical ventilation in intensive care unit Zheng, L., Xue, Y.-J., Yuan, Z.-N., Xing, X.-Z. Scientific Reports, 2025 https://www.scopus.com/pages/publications/105000536043?origin=scopusAI

125. Machine learning prediction models for mechanical ventilation requirement in patients with embolic stroke: a retrospective cohort study based on MIMIC-IV Ma, L., Liu, X., Nan, R., (...), Wang, X. Frontiers in Medicine, 2026 https://www.scopus.com/pages/publications/105051501546?origin=scopusAI

126. Development and validation of a mechanical power-oriented nomogram model for predicting the risk of weaning failure in mechanically ventilated patients: an analysis using the data from MIMIC-IV Yan, Y., Xie, Y., Luo, J., (...), Li, X. Chinese Critical Care Medicine, 2023 https://www.scopus.com/pages/publications/85166778542?origin=scopusAI

127. An explainable artificial intelligence framework for weaning outcomes prediction using features from electrical impedance tomography Wang, P., Chen, T.-H., Chang, M.-Y., (...), Zhao, Z. Computer Methods and Programs in Biomedicine, 2025 https://www.scopus.com/pages/publications/105004401237?origin=scopusAI

128. Early prediction of prolonged mechanical ventilation in invasively ventilated patients with subarachnoid hemorrhage: A MIMIC-IV prediction model study Rao, D., Li, B., Dai, X., (...), Liu, J. Clinical Neurology and Neurosurgery, 2026 https://www.scopus.com/pages/publications/105042599909?origin=scopusAI

129. Development and validation of a machine learning model for real-time prediction of invasive mechanical ventilation weaning readiness Zappalà, S., Scaravilli, V., Rovati, L., (...), Grasselli, G. Journal of Critical Care, 2025 https://www.scopus.com/pages/publications/105005956064?origin=scopusAI

130. Operationalization of Artificial Intelligence Applications in the Intensive Care Unit A Systematic Review Berkhout, W.E.M., van Wijngaarden, J.J., Workum, J.D., (...), van Genderen, M.E. JAMA Network Open, 2025 https://www.scopus.com/pages/publications/105012229767?origin=scopusAI

131. Implementation frameworks for end-to-end clinical AI: Derivation of the SALIENT framework Van Der Vegt, A.H., Scott, I.A., Dermawan, K., (...), Lane, P.J. Journal of the American Medical Informatics Association, 2023 https://www.scopus.com/pages/publications/85168254443?origin=scopusAI

132. Evaluating machine learning models for sepsis prediction: A systematic review of methodologies Deng, H.-F., Sun, M.-W., Wang, Y., (...), Jiang, H. iScience, 2022 https://www.scopus.com/pages/publications/85122353912?origin=scopusAI

133. Consolidated Health Economic Evaluation Reporting Standards for Interventions That Use Artificial Intelligence (CHEERS-AI) Elvidge, J., Hawksworth, C., Avşar, T.S., (...), Wilson, E. Value in Health, 2024 https://www.scopus.com/pages/publications/85196424591?origin=scopusAI

134. Comparison of artificial intelligence and logistic regression models for mortality prediction in acute respiratory distress syndrome: a systematic review and meta-analysis He, Y., Liu, N., Yang, J., (...), Zhang, Z. Intensive Care Medicine Experimental, 2025 https://www.scopus.com/pages/publications/85218672657?origin=scopusAI

135. Artificial Intelligence for Predicting Difficult Airways: A Review Alatau, M., Bauer, J., Sazonov, V. Journal of Clinical Medicine, 2025 https://www.scopus.com/pages/publications/105024579526?origin=scopusAI

136. An artificial intelligence application to predict prolonged dependence on mechanical ventilation among patients with critical orthopaedic trauma: an establishment and validation study Jiang, W., Liu, T., Sun, B., (...), Lei, M. BMC Musculoskeletal Disorders, 2024 https://www.scopus.com/pages/publications/85213682570?origin=scopusAI

137. External Validation of the Augmented Renal Clearance Predictor in Critically Ill COVID-19 Patients Huang, C.-Y., Güiza, F., Gijsen, M., (...), De Vlieger, G. Antibiotics, 2023 https://www.scopus.com/pages/publications/85153753594?origin=scopusAI

138. Machine learning models to predict ICU admission of patients with COPD and depression: A retrospective cohort study Chen, H., Guo, Y., You, T., Shi, K. Medicine (United States), 2026 https://www.scopus.com/pages/publications/105043878344?origin=scopusAI

139. Artificial intelligence models for predicting pulmonary complications in patients with chest trauma: a retrospective study Seok, J., Lee, J., Kang, W.S. Journal of Trauma and Injury, 2025 https://www.scopus.com/pages/publications/105018046685?origin=scopusAI

140. RMS: a ML-based system for ICU respiratory monitoring and resource planning Hüser, M., Lyu, X., Faltys, M., (...), Rätsch, G. npj Digital Medicine, 2025 https://www.scopus.com/pages/publications/105025454038?origin=scopusAI

141. Correlation between oxygenation function and laboratory indicators in COVID-19 patients based on non-enhanced chest CT images and construction of an artificial intelligence prediction model Kong, W., Liu, Y., Li, W., (...), Jiao, G. Frontiers in Microbiology, 2024 https://www.scopus.com/pages/publications/85209588349?origin=scopusAI

142. Evaluating the efficacy of large language models in predicting intensive care unit admission needs Turan, E.İ., Baydemir, A.E., Turan, Z.P., Şahin, A.S. Journal of Critical Care, 2026 https://www.scopus.com/pages/publications/105028270048?origin=scopusAI

143. Systematic review and meta-analysis of artificial intelligence models for diagnosing and subphenotyping ARDS in adults Muñoz, J., Ruíz-Cacho, R., Fernández-Araujo, N.J., (...), Muñoz-Visedo, J. Heart and Lung, 2026 https://www.scopus.com/pages/publications/105017442082?origin=scopusAI

144. Latent class analysis-derived subphenotypes are generalisable to observational cohorts of acute respiratory distress syndrome: A prospective study Sinha, P., Delucchi, K.L., Chen, Y., (...), Calfee, C.S. Thorax, 2022 https://www.scopus.com/pages/publications/85110189773?origin=scopusAI

145. Prediction of ICU needs and organ failures in infectious patients using machine learning Ma, C., Xu, R., Su, D., (...), Ning, W. iScience, 2025 https://www.scopus.com/pages/publications/105024193027?origin=scopusAI

146. Development and validation of a score for prediction of postoperative respiratory complications in infants and children (SPORC-C) Luedeke, C.M., Rudolph, M.I., Pulverenti, T.S., (...), Eikermann, M. British Journal of Anaesthesia, 2025 https://www.scopus.com/pages/publications/85200595922?origin=scopusAI

147. How Few Sensors Are Enough? Minimum Vital Sign Requirements for ML-Based Deterioration Prediction Myroshnyk, Y., Leshchenko, O. IEEE Access, 2026 https://www.scopus.com/pages/publications/105050193839?origin=scopusAI

148. Value of laboratory results in addition to vital signs in a machine learning algorithm to predict in-hospital cardiac arrest: A singlecenter retrospective cohort study Ueno, R., Xu, L., Uegami, W., (...), Jones, D. PLoS ONE, 2020 https://www.scopus.com/pages/publications/85088045954?origin=scopusAI

149. Early prediction of sepsis using an XGBoost model with single time-point non-invasive vital signs and its correlation with C-reactive protein and procalcitonin: A multi-center study Yang, A.C., Ma, W.-M., Chiang, D.-H., (...), Wang, C.-Y. Intelligence-Based Medicine, 2025 https://www.scopus.com/pages/publications/105001281788?origin=scopusAI

150. Predictive Model of Dynamic Subphenotypes for 30-Day Mortality in Emergency Department Patients with Suspected Infection Using the Vital Signs of the First 24 Hours: An Analytical Cohort Study in a Tertiary Care Clinic Patiño-Moncayo, A., Ibañez-Pinilla, M., Gonzalez-Varela, M., (...), Patiño-Unibio, L.F. Journal of Clinical Medicine, 2026 https://www.scopus.com/pages/publications/105033957814?origin=scopusAI

151. Comparing machine learning algorithms for predicting acute kidney injury Parreco, J., Soe-Lin, H., Parks, J.J., (...), Rattan, R. American Surgeon, 2019 https://www.scopus.com/pages/publications/85071282243?origin=scopusAI

152. A Machine Learning Approach for Predicting In-Hospital Cardiac Arrest Using Single-Day Vital Signs, Laboratory Test Results, and International Classification of Disease-10 Block for Diagnosis Park, H., Park, C.S. Annals of Laboratory Medicine, 2025 https://www.scopus.com/pages/publications/85216351481?origin=scopusAI

153. Towards Reliable ARDS Clinical Decision Support: ARDS Patient Analytics with Free-text and Structured EMR Data Apostolova, E., Uppal, A., Galarraga, J.E., (...), Velez, T. AMIA ... Annual Symposium proceedings. AMIA Symposium, 2019 https://www.scopus.com/pages/publications/85083755322?origin=scopusAI

154. Added value of inflammatory markers to vital signs for predicting mortality in patients with suspected infection: external validation and model development Takada, T., Hoogland, J., Fujii, K., (...), Fukuhara, S. Internal and Emergency Medicine, 2025 https://www.scopus.com/pages/publications/85210086029?origin=scopusAI

155. Early warning score adjusted for age to predict the composite outcome of mortality, cardiac arrest or unplanned intensive care unit admission using observational vital-sign data: A multicentre development and validation Shamout, F., Zhu, T., Clifton, L., (...), Clifton, D.A. BMJ Open, 2019 https://www.scopus.com/pages/publications/85075324627?origin=scopusAI

156. Artificial intelligence in intensive care medicine Baumgart, A., Beck, G., Ghezel-Ahmadi, D. Medizinische Klinik - Intensivmedizin und Notfallmedizin, 2024 https://www.scopus.com/pages/publications/85189167492?origin=scopusAI

157. Artificial intelligence in intensive care units: a scoping review addressing the translational gap to clinical practice Zarantonello, F., De Cassai, A., Pettenuzzo, T., (...), Boscolo, A. Korean Journal of Anesthesiology, 2026 https://www.scopus.com/pages/publications/105040167466?origin=scopusAI

158. Harnessing AI in critical care: opportunities, challenges and key steps for success Kalimouttou, A., Stevens, R.D., Pirracchio, R. Thorax, 2026 https://www.scopus.com/pages/publications/105008968911?origin=scopusAI

159. Use of artificial intelligence in critical care: opportunities and obstacles Pinsky, M.R., Bedoya, A., Bihorac, A., (...), Clermont, G. Critical Care, 2024 https://www.scopus.com/pages/publications/85189944849?origin=scopusAI

160. Artificial intelligence in clinical decision support and the prediction of adverse events Oei, S.P., Bakkes, T.H.G.F., Mischi, M., (...), Turco, S. Frontiers in Digital Health, 2025 https://www.scopus.com/pages/publications/105008074188?origin=scopusAI

161. Machine Learning and Artificial Intelligence in Intensive Care Medicine: Critical Recalibrations from Rule-Based Systems to Frontier Models Hadweh, P., Niset, A., Salvagno, M., (...), Barrit, S. Journal of Clinical Medicine, 2025 https://www.scopus.com/pages/publications/105009146349?origin=scopusAI

162. Artificial intelligence for clinical decision support for monitoring patients in cardiovascular ICUs: A systematic review Moazemi, S., Vahdati, S., Li, J., (...), Schmid, F. Frontiers in Medicine, 2023 https://www.scopus.com/pages/publications/85153408357?origin=scopusAI

163. An Explanation User Interface for Artificial Intelligence–Supported Mechanical Ventilation Optimization for Clinicians: User-Centered Design and Formative Usability Study Jung, I.-C., Zerlik, M., Schuler, K., (...), Sedlmayr, B. JMIR Formative Research, 2026 https://www.scopus.com/pages/publications/105029983024?origin=scopusAI

164. Evaluating the impact of explainable AI on clinicians' decision-making: A study on ICU length of stay prediction Jung, J., Kang, S., Choi, J., (...), Kim, H. International Journal of Medical Informatics, 2025 https://www.scopus.com/pages/publications/105004061746?origin=scopusAI

165. Explainable AI for Clinical Decision Support Systems: Literature Review, Key Gaps, and Research Synthesis Salimparsa, M., Sedig, K., Lizotte, D.J., (...), Muanda, F.T. Informatics, 2025 https://www.scopus.com/pages/publications/105025920568?origin=scopusAI

166. Decoding the "black-box": explainable artificial intelligence towards trustworthy advancement in respiratory medicine Marchi, G. Breathe, 2026 https://www.scopus.com/pages/publications/105039564257?origin=scopusAI

167. Towards Trustful Machine Learning for Antimicrobial Therapy Using an Explainable Artificial Intelligence Dashboard De Corte, T., Verhaeghe, J., Ongenae, F., (...), Van Hoecke, S. Applied Sciences (Switzerland), 2025 https://www.scopus.com/pages/publications/105020244942?origin=scopusAI

168. Exploring the Ethical Dimensions of AI-Driven Patient Monitoring in Critical Care Nursing: A Systematic Review Amin, S.M., Almagharbeh, W.T., Alrimawi, I., (...), El-sayed, M.M. Nursing in Critical Care, 2026 https://www.scopus.com/pages/publications/105029613531?origin=scopusAI

169. Leading with AI in critical care nursing: challenges, opportunities, and the human factor Hassan, E.A., El-Ashry, A.M. BMC Nursing, 2024 https://www.scopus.com/pages/publications/85206386246?origin=scopusAI

170. Implementing Artificial Intelligence in Critical Care Medicine: a consensus of 22 Cecconi, M., Greco, M., Shickel, B., (...), Bihorac, A. Critical Care, 2025 https://www.scopus.com/pages/publications/105010214794?origin=scopusAI

171. AI ethics for the everyday intensivist Tan, S.C., Modra, L., Hensman, T. Critical Care and Resuscitation, 2025 https://www.scopus.com/pages/publications/105009006235?origin=scopusAI

172. Challenges in Health Equity: Balancing Innovation With Fairness and Addressing Algorithmic Biases Arshed, M., Kazi, A.K., Kiran, M., Ali, S.M.F. Innovation and Transformation of Public Health, 2025 https://www.scopus.com/pages/publications/105018998360?origin=scopusAI

173. Artificial Intelligence in Critical Care Nursing: Benefits, Risks, and Ethical Considerations George, A., Peirce, A.G. Critical Care Nurse, 2025 https://www.scopus.com/pages/publications/105017695572?origin=scopusAI

174. Artificial intelligence in healthcare: Tailoring education to meet EU AI-Act standards Bignami, E., Darhour, L.J., Buhre, W., (...), Bellini, V. Health Policy and Technology, 2025 https://www.scopus.com/pages/publications/105011830612?origin=scopusAI

175. The Role of Machine Learning and Artificial Intelligence in Enhancing Critical Care Nursing Practice: A Scoping Review Alqaisi, O., Al-Ghabeesh, S., Dibas, M., (...), Tai, P. Nursing in Critical Care, 2026 https://www.scopus.com/pages/publications/105048168237?origin=scopusAI

176. Perceptions on artificial intelligence among anaesthesia and intensive care professionals: An international survey on attitudes, expectations and needs Theilen, R., Kramer, T., Scharffenberg, M., (...), Wittenstein, J. Journal of Clinical Monitoring and Computing, 2026 https://www.scopus.com/pages/publications/105043765827?origin=scopusAI

177. From Code to Clinic: Bridging the Implementation Gap in AI-Based Clinical Decision Support Savadkoohi, E.G., Asadi, F., Emami, H. 2026 International Interdisciplinary Conference on Artificial Intelligence: Engineering, Health, Finance and Humanities, IICAI 2026, 2026 https://www.scopus.com/pages/publications/105046974656?origin=scopusAI

178. Multicenter derivation and validation of an early warning score for acute respiratory failure or death in the hospital Dziadzko, M.A., Novotny, P.J., Sloan, J., (...), Gong, M.N. Critical Care, 2018 https://www.scopus.com/pages/publications/85055615148?origin=scopusAI

179. A clinically interpretable prediction model for acute mortality in patients with pneumonia requiring mechanical ventilation Hu, H., Zu, Y., Zhao, L., (...), Chang, D. Respiratory Research, 2026 https://www.scopus.com/pages/publications/105033507420?origin=scopusAI

180. Artificial Intelligence in the Management of Patients with Respiratory Failure Requiring Mechanical Ventilation: A Scoping Review Viderman, D., Ayazbay, A., Kalzhan, B., (...), Abdildin, Y. Journal of Clinical Medicine, 2024 https://www.scopus.com/pages/publications/85213241877?origin=scopusAI

181. Mean airway pressure as an independent risk factor for developing acute kidney injury in mechanically ventilated critically ill patients: a multi-center retrospective analysis Gao, Y., Zhang, J., Xue, X., (...), Xie, K. Renal Failure, 2026 https://www.scopus.com/pages/publications/105040046668?origin=scopusAI

182. Use of Machine Learning to Screen for Acute Respiratory Distress Syndrome Using Raw Ventilator Waveform Data Rehm, G.B., Cortés-Puch, I., Kuhn, B.T., (...), Adams, J.Y. Critical Care Explorations, 2021 https://www.scopus.com/pages/publications/85140739896?origin=scopusAI

183. Predicting Tracheostomy Need on Admission to the Intensive Care Unit—A Multicenter Machine Learning Analysis Nguyen, M., Amanian, A., Wei, M., (...), Mendez-Tellez, P.A. Otolaryngology - Head and Neck Surgery (United States), 2024 https://www.scopus.com/pages/publications/85200036439?origin=scopusAI

184. Guidance for reporting artificial intelligence technology evaluations for ultrasound scanning in regional anaesthesia (GRAITE-USRA): an international multidisciplinary consensus reporting framework Zhang, X., Ferry, J., Hewson, D.W., (...), Graff, V. Anaesthesia, 2025 https://www.scopus.com/pages/publications/105018100359?origin=scopusAI

185. Moving towards vertically integrated artificial intelligence development Zhang, J., Budhdeo, S., William, W., (...), Teo, J.T. npj Digital Medicine, 2022 https://www.scopus.com/pages/publications/85139225808?origin=scopusAI

186. Artificial intelligence for prediction of treatment outcomes in breast cancer: Systematic review of design, reporting standards, and bias Corti, C., Cobanaj, M., Marian, F., (...), Curigliano, G. Cancer Treatment Reviews, 2022 https://www.scopus.com/pages/publications/85132454360?origin=scopusAI

187. Reporting Standards for Diagnostic Testing: Guidance for Authors From Editors of Respiratory, Sleep, and Critical Care Journals Ost, D.E., Feller-Kopman, D.J., Gonzalez, A.V., (...), Smyth, A. Journal of Bronchology and Interventional Pulmonology, 2023 https://www.scopus.com/pages/publications/85164210093?origin=scopusAI

188. Implementation and evaluation of a 24/7 virtual intensive care support model in regional Australia: A controlled quasi-experimental study Bhonagiri, D., Hillman, K., Lakshmanan, R., (...), Andersen, J. Australian Critical Care, 2026 https://www.scopus.com/pages/publications/105044962619?origin=scopusAI

189. Assessing the quality of reporting in artificial intelligence/machine learning research for cardiac amyloidosis Arefeen, A., Singh, S., Razavi, C., (...), Dev, S. JAMIA Open, 2025 https://www.scopus.com/pages/publications/105018212249?origin=scopusAI

190. Design and Implementation of an Intensive Care Unit Command Center for Medical Data Fusion Feng, W.-S., Chen, W.-C., Lin, J.-Y., (...), Lin, Y.-B. Sensors, 2024 https://www.scopus.com/pages/publications/85197175092?origin=scopusAI